Lambda-1 Detector: Adaptive Interference Detection in Synthetic Aperture Radar Images

Huizhang Yang, *Member, IEEE*, Ping Lang, *Member, IEEE*, Yaomin He, Xingyu Lu, *Member, IEEE*, Zhong Liu, *Member, IEEE*, and Jian Yang, *Senior Member, IEEE*

***Abstract*— This article proposes a novel eigenvalue-based detector, called Lambda-1 detector, for adaptive and robust interference detection in single-look-complex (SLC) synthetic aperture radar (SAR) images. The proposed method leverages the increased eigenvalues caused by interference in SAR image blocks, where the interference is expected to have a small set of eigenvalues, particularly with a dominating one. Specifically, the method segments the image into multiple blocks, computes the eigenvalues of each block's covariance matrix, and compares the largest eigenvalue λ_1 with a threshold to determine the presence of interference under the criteria of constant false alarm rate (CFAR), thereby enabling adaptive interference detection against varying levels of interference-to-signal ratios (ISRs). The largest eigenvalue is characterized by the order-2 Tracy–Widom distribution (no closed-form expression) under the assumption of the image's homogeneity, and the threshold is adaptively determined based on a scaled and shifted Gamma distribution that fits this distribution with a closed-form expression. The method is robust by first modeling and then correcting the impacts of upsampling and windowing of SAR image data on the fit distribution's parameters, and by incorporating outlier removal preprocessing. Experimental results validate the effectiveness of the proposed method in successfully detecting both strong and weak interferences in various SAR images, including Sentinel-1 and Gaofen-3. The detection performance is quantitatively evaluated using false alarm rate P_{fa} and detection rate P_d. In summary, the proposed Lambda-1 detector effectively identifies interference artifacts in focused SAR images and holds promise for improving the quality of SAR imagery by incorporating adaptive interference removal.**

***Index Terms*— Constant false alarm rate (CFAR), interference detection, radio interference, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) system utilizes electromagnetic waves to capture radar images of the Earth's surface and is the foundation of many remote sensing applications. However, the quality of SAR data can be severely degraded by signal interference arising from various radio sources, such as communication systems, radar systems, and other electronic devices. Fig. 1 shows some example images corrupted by various interference signals. It is clear that the impact of interference on SAR data is significant, leading to reduced image quality.

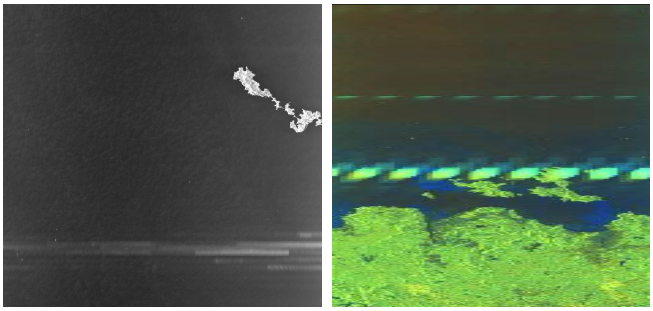


Fig. 1. Examples of SAR images containing interference.

A. Interference Sources, Types, and Impacts

Interference signals in SAR imagery originate from a variety of sources, including terrestrial and spaceborne ones. Terrestrial sources include commercial and military radio transmissions, such as FM/TV broadcasts and radar systems, which emit signals that can interfere with SAR systems operating in similar frequency bands. Similarly, spaceborne sources like communication satellites or other radar satellites sharing the same frequency band can cause interference.

The types of interference signals can be categorized into narrowband, pulse-type wideband, and continuous-wave wideband ones [1]. Narrowband interference has a relatively small bandwidth and is continuous over the duration of the SAR pulse, while pulse-type wideband interference has a larger bandwidth but appears intermittently over the pulse repetition period. Continuous-wave wideband interference has a broad bandwidth and is continuous over the SAR pulse duration, resembling wideband noise.

The impact of interference on image quality is substantial, including reduced signal-to-noise ratio, distorted dynamic range, defocused images, amplitude and phase distortions, and degraded spatial resolution. These behaviors will finally lead to reduced accuracy of information retrieval in subsequent applications. In short, interference significantly degrades SAR image quality and introduces errors in subsequent applications,

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Huizhang Yang, Xingyu Lu, and Zhong Liu are with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: hzyang@njust.edu.cn).

Ping Lang, Yaomin He, and Jian Yang are with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: plang1220@sina.com).

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emphasizing the importance of developing robust interference detection and mitigation techniques.

B. Existing Works on Interference Detection in SAR Data

The detection and mitigation of interference in SAR data have become critical research topics in the field of remote sensing. This article focuses on the detection stage. Numerous methods have been proposed for interference detection in SAR data, which can be broadly categorized as follows.

Traditional signal processing-based methods [2], [3], [4], [5], [6], [7], [8], [9], [10], [11]: many works perform interference detection using processing flows consisting of traditional signal processing methods. These methods first extract interference features in certain domains (such as frequency domain [2], [4], [5], [11], multichannel domain [6], [8], and eigenvalue domain [7], [9], [10]) via tools, such as spectral analysis and eigenvalue decomposition. Then, they perform interference detection using these features via statistical tests, empirical thresholding, or clarification methods such as k-means clustering. For example, Yang et al. [2] used the range frequency feature of interference (i.e., interference will raise the power of certain frequencies) and proposed a subband spectral kurtosis detector to detect interference. Natsuaki et al. [3] used the azimuth frequency feature of intermittently transmitted interferences and proposed an autocorrelation-based detection method for this type of interference in the azimuth-frequency and range-time domain of SAR images. Monti-Guarnieri et al. [4] used the range spectrum feature of interference and performed interference detection using a combination of the one-tailed Fisher's Z test on the echo range spectrum. Bollian et al. [5] also used this feature to perform interference detection via thresholding in their work. Li et al. [11] used a range spectral difference measure to capture interference features and then adopted k-means clustering to detect interference. Lv et al. [7] proposed a two-step method for detecting pulse interference using eigenvalue decomposition and the short-time Fourier transform. A common property of these works is that they perform interference detection in SAR raw data rather than focused images. Among these works, range frequency-domain detection methods are not suitable for wideband interference. Eigenvalue analysis-based methods can detect both wideband and narrowband interference provided that the interference data matrix has a low-rank structure. However, the extension of these works to interference detection in SAR images is not clear, and current works also lack statistical modeling.

Deep learning and neural networks based methods [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23]: Leveraging the progress of efficient deep neural networks in computer vision, these methods employ neural networks to learn features of interference from data and detect interference automatically. They can detect and segment interference from SAR data without manually designed or selected features of interferences, offering robustness against various interference types. These methods can use either SAR raw data or focused images. For the former case, the raw data are often preprocessed using a short-time Fourier transform to form spectrogram images and then fed to neural network detectors. It is also notable that some of these works perform interference detection and mitigation sequentially. For example, Yu et al. [12] proposed a multiclass interference detection method based on the single-shot multibox detector network. Fan et al. [13] proposed an interference detection method based on the deep residual network. Artiemjew et al. [14] used an LeNet-type convolutional neural network for interference detection. Tao et al. [15] proposed an approach for detecting interference signals in SAR raw data using a U-Net convolutional neural network. Chojka et al. [23] proposed a neural network detector using feature extraction involving pixel convolution, thresholding, and nearest neighbor structure filtering, which can effectively detect interference in sentinel-1 quick-look images. Despite that deep learning-based methods are promising in SAR interference detection, the training of the neural networks requires a large amount of labeled data, and the generalization ability of these networks to different SAR data sources may be a problem. The significantly increased computational burden of these works compared with the range frequency-domain methods is also a notable limitation.

Time-series analysis and change detection based methods [24], [25]: these methods focus on identifying changes in time-series SAR images that may indicate the presence of interference. They can be particularly effective for detecting unexpected or nonrepetitive interference sources. The background modeling and change detection for interference detection by Li et al. [24] and time-series analysis on Sentinel-1 SAR data for interference detection by Tao et al. [25] are examples of this approach. The requirement of multiple repeat-pass SAR data limits the application of these methods to single-pass SAR data.

C. Contributions of This Article

In this article, we introduce a novel interference detection approach, i.e., Lambda-1 detector, for effectively detecting interference artifacts in focused SAR images. The detection results can be used for the following applications: 1) spatial adaptive suppression of interference; 2) image quality assessment (e.g., how many areas are corrupted by interference; 3) coarse localization of interference source on the Earth; and 4) report of illegal microwave radiation reserved for SAR remote sensing. The core idea of our method is to leverage the eigenvalue analysis of the image block by block, which captures the increased eigenvalue caused by interference. Specifically, we analyze the eigenvalue distribution of image blocks, and find that many interference artifacts in the image domain are expected to have a small set of eigenvalues, particularly with a dominating largest eigenvalue. Using this property, our method first segments the image into multiple blocks, computes the eigenvalues of each block's covariance matrix, and then compares the largest eigenvalue to a threshold to determine the presence of interference. The threshold is adaptively determined under the constant false alarm rate (CFAR) criteria based on the distribution of the largest eigenvalue λ_1 under the assumption of the image's homogeneity, where λ_1 follows the order-2 Tracy–Widom distribution. The

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adaptive threshold allows the method to be robust against varying levels of interference-to-signal ratios (ISRs). The method is also shown to be robust against the presence of outliers through a preprocessing step that masks out detected outliers. Considering that the proposed method performs interference detection by comparing the largest eigenvalue λ_1 with the threshold, we name our detector as Lambda-1 detector.

Experimental results validate the effectiveness of the proposed method. Our method is demonstrated to successfully detect both strong and weak interference in various SAR images, including Sentinel-1 and Gaofen-3. The detection performance is quantitatively evaluated using metrics, such as false alarm rate (P_{fa}) and detection rate (P_d).

The contributions of this article are summarized as follows.

1) *Lambda-1 Detector Framework:* We establish an interference detection framework by using the order-2 Tracy–Widom distribution of the largest eigenvalue for homogeneous images, enabling the selection of a CFAR threshold for interference detection. Compared with neural network-based methods, our detector does not require training data. The detector is a spatial detector, which can be used to find the interference-corrupted area of single-look-complex (SLC) images. The detection results can be used for image quality evaluation, and support the automatic down-stream processing workflow by adaptive interference removal.
2) *Oversampling and Windowing Correction:* We address the impact of upsampling and windowing on the distribution's parameters, and introduce a threshold correction scheme to enhance detection robustness.
3) *Integrated Block Adaptive Processing:* We propose an integrated block adaptive processing framework that consists of interference detection and mitigation, allowing for joint detection and removal of interference.

Our code is online available at https://github.com/huizhangyang/lambda1-bsf.

The remainder of this article is organized as follows. Section II presents the signal model for interference in SAR images. Section III details the eigenvalue CFAR detection method. Section IV discusses the threshold selection for upsampled and windowed data. Section V performs the implementation details and complexity analysis. Section VI provides the experimental results to validate the effectiveness of our approach. Section VII concludes this article.

II. Signal Model

In this section, we will introduce our signal model for radio interference observed in SAR SLC images. The model has been proven effective in characterizing many common interference signals observed in spaceborne SAR data. Under our model, we will show that the presence of interference will increase the eigenvalues of the image data, which forms the basis of our subsequent interference detection method detailed in Section IV.

A. Modeling Interference From Echo Domain to SLC Image Domain

We model radio interference observed in SAR data as the sum of several decoupled signal components in range and azimuth domains [26]. Specifically, the interference signal $y(\eta, t)$ can be expressed as a sum of L interference components

$$y(\eta, t) = \sum_{l=1}^{L} y_l(\eta, t) = \sum_{l=1}^{L} a_l(\eta) r_l(t) \tag{1}$$

where $a_l(\eta)$ and $r_l(t)$ are 1-D functions characterizing the l interference signal component in azimuth time η and range time t, respectively.

SAR focusing can be viewed as a 2-D matched filtering process using a time-varying kernel. The kernel can be expressed as $h(\eta, t; \eta', t')$, where (η', t') is the matched point in azimuth and range time. The interference response in the SLC image domain can be obtained by convolving the interference signal with this kernel

$$\tilde{y}(\eta', t') = \sum_{l=1}^{L} \int\int a_l(\eta) r_l(t) * h(\eta, t; \eta', t') d\eta dt. \tag{2}$$

Using order-N Taylor series approximation for the kernel

$$h(\eta, t; \eta', t') \approx \sum_{n=1}^{N} u_n(\eta - \eta') v_n(t - t'). \tag{3}$$

The interference response $\tilde{y}(\eta', t')$ can be further simplified using the following approximated signal model:

$$\tilde{y}(\eta', t') \approx \sum_{l=1}^{L} \sum_{n=1}^{N} \tilde{a}_{l,n}(\eta') \cdot \tilde{r}_{l,n}(t') \tag{4}$$

where $\tilde{a}_{l,n}(\eta')$ is the convolutional output of $a_l(\eta)$ and $u_n(\eta)$, and $\tilde{r}_{l,n}(t')$ is the convolutional output of $r_l(t)$ and $v_n(t)$. This equation states that the interference response in the SLC image domain can be expressed as the summation of several range-azimuth decoupled signals. The interference responses exist in the image domain in form of bright radiometric artifacts. Examples of such interference artifact in SAR images can be found in Section VI. The detection of the presence and absence of interference can help to evaluate the quality of the acquired image, and the detection results can also be used to perform adaptive interference filtering.

B. Eigenvalue Analysis of SLC Images Containing Interference

The dimension of the interference subspace, which we denote by K, satisfies $K \leq \text{LN}$. In real SAR data, the interference components often can be approximated using a small K value. Then, the interference response in (4) can be decomposed as the summation of K decoupled signals. This decomposition allows us to re-express the interference response (4) using eigenvalues and eigenvectors

$$\tilde{y}(\eta', t') \approx \sum_{k=1}^{K} \sqrt{\lambda_k} \cdot p_k(\eta') \cdot q_k(t'). \tag{5}$$

In the discrete domain, this corresponds to the singular value decomposition (SVD), where $\sqrt{\lambda_k}$, $p_k(\eta')$, and $q_k(t')$ correspond to the singular values, left and right eigenvectors, respectively. λ_k, $k = 1, \ldots, K$ are the eigenvalues of the data covariance matrix arranged in the descending order.

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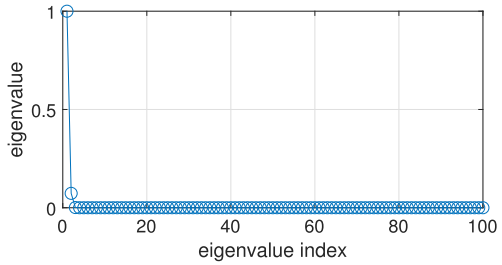


Fig. 2. Top 100 eigenvalues of a simulated LFM pulse interference in a stripmap SAR image with 2048 × 2048 pixels.

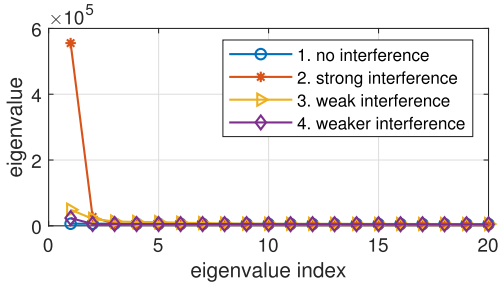


Fig. 3. Examples of eigenvalue distributions of SAR SLC image data with and without interferences. The top 20 eigenvalues are plotted. These eigenvalues are calculated from four Sentinel-1 images normalized to the same Frobenius norm. The first image does not have interference, and the rest three have strong (ISR = 0.96 dB), weak (ISR = −12.88 dB), and weaker interference (ISR = −16.12 dB), respectively.

According to this approximation, we can find that the energy of the interference component will be concentrated in a few eigenvalues particularly the first one. To show this point, we simulate an linear frequency modulation (LFM) pulse interference as a matrix $\mathbf{Y}$ in the SAR image domain, calculate the eigenvalues of its data covariance matrix $\mathbf{R} = \mathbf{Y}\mathbf{Y}^{\mathrm{H}}$ (here, $(\cdot)^{\mathrm{H}}$ refers to the Hermitian transpose), and then plot in Fig. 2 the top 100 eigenvalues. The figure clearly illustrates that the first eigenvalue is significantly larger than the others, confirming our hypothesis that interference has a dominating largest eigenvalue.

If interference exists, an SLC image denoted by a matrix $\mathbf{S}$, can be represented as the sum of a clean image and an interference component

$$\mathbf{S} = \mathbf{S}_{\text{clean}} + \mathbf{Y} \tag{6}$$

where $\mathbf{S}_{\text{clean}}$ is the underlying clean image. According to the previous analysis, the largest eigenvalue λ_1 of the image covariance matrix $\mathbf{R} = \mathbf{S}\mathbf{S}^{\mathrm{H}}$ will be significantly larger than that of a clean one if the ISR is not too weak. Fig. 3 shows some examples of the top 20 eigenvalues of a clean image and three interference-corrupted images, which confirms the above analysis. Based on this fact, we will proceed to develop a statistical detection method that exploits this discrepancy in eigenvalues.

III. Development of Lambda-1 Detector for Interference Detection in SAR Images via Statistical Modeling

In this section, we develop the framework of our eigenvalue CFAR method for adaptive interference detection in SAR images.

A. Interference Detection as a Hypothesis Test Problem

The detection of interference in SAR images can be conceptualized as a binary hypothesis testing process, where two scenarios are considered.

1) *H0:* The image is homogeneous, consisting only of background clutter and noise, without any interference artifacts.
2) *H1:* The image contains interference artifacts superimposed on the homogeneous background.

Under the assumption of homogeneity (H0), the eigenvalues of the image should ideally exhibit a stable behavior within certain predictable bounds. In order to perform detection under the criteria of CFAR, we will develop the distribution of the largest eigenvalue, based on which we can select a threshold of the largest eigenvalue under a given false alarm rate. Then, eigenvalues exceeding this threshold indicate the presence of interference artifacts under the criteria of CFAR. The detailed development is provided as follows.

B. Statistical Distribution of the Largest Eigenvalue of Image Covariance Matrix Under H0

Consider that each image block is an $N \times M$ matrix $\mathbf{S}$, and assume that each element is i.i.d. zero-mean complex Gaussian random variable with variance σ^2 (i.e., the image is homogeneous). It is notable that in practice SAR image pixels may not be statistically independent; however, we will ignore the correlation of pixels for theoretical simplicity. The covariance matrix $\mathbf{W} = \mathbf{S}\mathbf{S}^{\mathrm{H}}$ is an uncorrelated complex Wishart matrix. In the literature, it is shown that the probability density function (p.d.f.) of the largest eigenvalue of complex Wishart matrices can be expressed as a combination of Gamma distribution [27]. For a large dimension $N \times M$, the largest eigenvalue of $\mathbf{W} = \mathbf{S}\mathbf{S}^{\mathrm{H}}$ can be approximated by the Tracy–Widom distribution of order 2 (TW_2) and the p.d.f. is expressed as follows:

$$x \triangleq \frac{\lambda_1/\sigma^2 - a}{b} \sim \mathrm{TW}_2 \tag{7}$$

where the scaling and centering parameters a and b in the above expression are defined as follows:

$$\begin{aligned} a &\triangleq \left(\sqrt{N} + \sqrt{M}\right)^2 \\ b &\triangleq \sqrt{a}\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{M}}\right)^{1/3}. \end{aligned} \tag{8}$$

The cumulative distribution function (cdf) of the TW_2 random variable x can be expressed as follows:

$$F_{\mathrm{TW}_2}(x) = \exp\left\{-\int_x^{\infty} (y-x)q^2(y)dy\right\} \tag{9}$$

where $q(y)$ is the unique solution to the Painlevé II differential equation. The distribution does not have closed-form expression but can be evaluated numerically by solving the Painlevé II differential equation. Alternatively, the Tracy–Widom distribution can be accurately approximated by

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a scaled and shifted Gamma distribution with closed-form expression [27] as follows:

$$f_{\mathrm{TW}_2}(x;k,\theta,\alpha) \simeq \frac{1}{\Gamma(k)\theta^k}(x+\alpha)^{k-1}e^{-\frac{x+\alpha}{\theta}}, \quad x > -\alpha$$
$$F_{\mathrm{TW}_2}(x;k,\theta,\alpha) \simeq \frac{1}{\Gamma(k)}\gamma\left(k, \frac{x+\alpha}{\theta}\right), \quad x > -\alpha \tag{10}$$

where $\gamma(k,z) = \int_0^z t^{k-1}e^{-t}dt$ is the incomplete Gamma function, the parameters $k = 79.6595$, $\theta = 0.101037$, and $\alpha = 9.81961$ are obtained via fitting the first three moments (i.e., the mean, variance, and skewness) of the true TW_2 distribution using the scaled and shifted Gamma distribution.

C. Threshold Determination Under a Given False Alarm Rate P_{fa}

If the largest eigenvalue of the covariance matrix exceeds a sufficiently large threshold λ_{th}, we can judge that the image contains interference. The false alarm rate under H0 (i.e., mistakenly reporting the existence of interference) can be modeled as follows:

$$P_{\mathrm{fa}} = P(\lambda_1 > \lambda_{\mathrm{th}}) = 1 - F_{\mathrm{TW}_2}\left(\frac{\lambda_{\mathrm{th}}/\sigma^2 - a}{b}\right). \tag{11}$$

Based on this equation, we can derive that the threshold x_{th} is related to the false alarm rate via the TW_2 distribution as follows:

$$\lambda_{\mathrm{th}} = \left(\left(F^{-1}_{\mathrm{TW}_2}(1-P_{\mathrm{fa}};k,\theta,\alpha)-\alpha\right)b - a\right)\sigma^2. \tag{12}$$

Here, $F^{-1}_{\mathrm{TW}_2}(x;k,\theta,\alpha)$ denotes the inverse function of $F_{\mathrm{TW}_2}(x;k,\theta,\alpha)$.

In practice, the variance parameter σ^2 is not a known parameter and thus should be estimated from data. To this end, we use the following unbiased estimator:

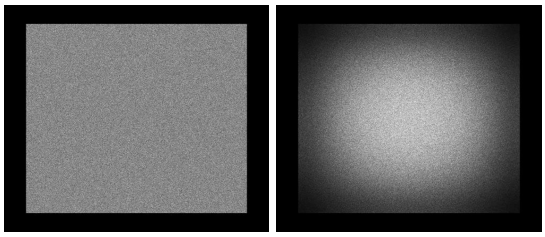
$$\sigma^2 \approx \hat{\sigma}^2 \triangleq \frac{1}{NM-1}\sum_{n=1,m=1}^{N,M}|S[n,m]|^2. \tag{13}$$

The variance is a measure of the image intensity level, and the threshold in (12) is adaptive to this quantity.

Because the proposed method performs interference detection by comparing the largest eigenvalue λ_1 with the CFAR threshold, we name our detector as Lambda-1 detector.

IV. Threshold Selection With Upsampled and Colored Image Data for Lambda-1 Detector

The eigenvalue threshold selection method established in the previous section is based on the assumption that SAR image pixels follow an independent and identical Gaussian distribution. Under this assumption, the statistical properties of the homogeneous image resemble those of Gaussian white noise. The Gaussian distribution is easily satisfied for homogeneous scenes. However, in practice, due to upsampling and windowing (as illustrated in Fig. 4), the pixels in the image typically do not satisfy independence. This phenomenon results in increased levels of the largest eigenvalue. Consequently, it is necessary to appropriately increase the eigenvalue threshold for the binary hypothesis test; otherwise, it will lead to an unacceptable high false alarm rate.



(a) (b)

Fig. 4. Illustrative image spectrum with upsampling and windowing. (a) With upsampling and (b) with upsampling and windowing.

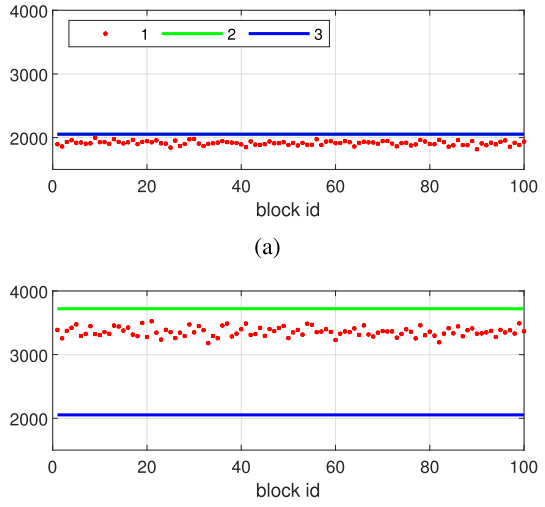


Fig. 5. Impacts of upsampling and windowing to the largest eigenvalues. Upsampling and windowing will raise the level of the largest eigenvalues. Data 1 (red point): the largest eigenvalues of 100 simulated Gaussian images, data 2 (green line): the corrected thresholds (its expression will be developed in Section IV), data 3 (blue line): the uncorrected thresholds. (a) Without upsampling and windowing and (b) with upsampling and windowing.

A. Influence of Upsampling on the Largest Eigenvalues

The distribution of the largest eigenvalue established in Section III depends on the sample variance and data dimensionality. Oversampling and windowing cause the distribution of the largest eigenvalue to deviate from the TW_2 distribution under the aforementioned ideal conditions, raising the levels of the largest eigenvalues as illustrated in Fig. 5. Therefore, it is essential to correct the TW_2 distribution parameters to make it closer to the actual distribution, and accordingly, the largest eigenvalue threshold should be selected for detection based on the corrected distribution.

Assuming that the observed scene is homogeneous, we will separately analyze the impacts of upsampling and windowing on the distribution of the largest eigenvalue. First, let us examine the factor of upsampling. An upsampled image can be represented by padding zeros in the frequency domain to the spectrum of the nonupsampled image. The corresponding upsampled image spectrum can be expressed by the following matrix:

$$\mathbf{S} = \begin{pmatrix} \mathbf{0}_{nm} & \mathbf{0}_{nq} & \mathbf{0}_{nm} \\ \mathbf{0}_{pm} & \tilde{\mathbf{S}} & \mathbf{0}_{pm} \\ \mathbf{0}_{nm} & \mathbf{0}_{nq} & \mathbf{0}_{nm} \end{pmatrix}. \tag{14}$$

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The image is related to the spectrum matrix via DFT, i.e., $\mathbf{I} = \mathbf{WSW}^{\mathrm{H}}$. Here, $\mathbf{W}$ is a normalized DFT matrix. The SVD of the discrete Fourier transform of the nonupsampled image can be expressed as $\tilde{\mathbf{S}} = \mathbf{UDV}$. Then, the following equation holds:

$$\mathbf{S} = \begin{pmatrix} \mathbf{0}_{nn} \\ \mathbf{U} \\ \mathbf{0}_{nn} \end{pmatrix} \mathbf{D}\begin{pmatrix} \mathbf{0}_{mm} & \mathbf{V} & \mathbf{0}_{mm} \end{pmatrix} = \tilde{\mathbf{U}}\mathbf{D}\tilde{\mathbf{V}}. \tag{15}$$

From (15), it can be observed that the eigenvalues of the frequency-domain matrix of the upsampled data matrix are the same as those of the frequency-domain matrix of the nonupsampled matrix. In the discrete domain, the Fourier transform can be implemented using a normalized discrete Fourier transform matrix, and this transformation does not alter the eigenvalues of the matrix. Therefore, for sampling matrices with the same Frobenius norm, the eigenvalues of the upsampled data matrix are identical to the eigenvalues of the frequency-domain matrix of their nonupsampled version. Consequently, the distribution of the largest eigenvalue of the upsampled image can be derived from this frequency-domain matrix. The azimuth length of this frequency-domain matrix is $L_1 = N_1 B_1 / f_{s1}$ and $L_2 = N_2 B_2 / f_{s2}$, and the variance of each element is $\sigma_{\tilde{\mathbf{S}}}^2 = \|\mathbf{I}\|_F^2/(N_1 N_2 - 1)$. Thus, the expression for the distribution of the largest eigenvalue is given by

$$x \triangleq \frac{\lambda_1/\sigma_{\tilde{\mathbf{S}}}^2 - c}{d} \sim \mathrm{TW}_2 \tag{16}$$

where

$$\begin{aligned} c &\triangleq \left(\sqrt{L_1} + \sqrt{L_2}\right)^2 \\ d &\triangleq \sqrt{c}\left(\frac{1}{\sqrt{L_1}} + \frac{1}{\sqrt{L_2}}\right)^{1/3}. \end{aligned} \tag{17}$$

The threshold for the largest eigenvalue derived from the above p.d.f. can eliminate the increased false alarm rate caused by upsampling.

B. Influence of Spectral Windowing on the Largest Eigenvalues

Next, we discuss the influence of spectral windowing on the distribution of the largest eigenvalue. We propose two methods to achieve this goal. The first method is to whitening the image, which means multiplying the image spectrum by the inverse of the window function. After this processing, the distribution of the largest eigenvalue of the image remains the same as that without windowing. Whitening requires a 2-D Fourier transform of the image and elementwise multiplication, which requires extra computational steps.

The second method treats the windowed spectrum in the frequency domain as an equivalent matrix $\tilde{\mathbf{S}}_{\mathrm{eq}}$ with a narrower equivalent bandwidth but without windows, as illustrated in Fig. 6. The equivalent bandwidths in azimuth and range are selected as follows:

$$\begin{aligned} B_{1,\mathrm{eq}} &= \frac{\int |W_1(f_1)|^2 df}{\max(|W_1(f_1)|^2)} \\ B_{2,\mathrm{eq}} &= \frac{\int |W_2(f_2)|^2 df}{\max(|W_2(f_2)|^2)}. \end{aligned} \tag{18}$$

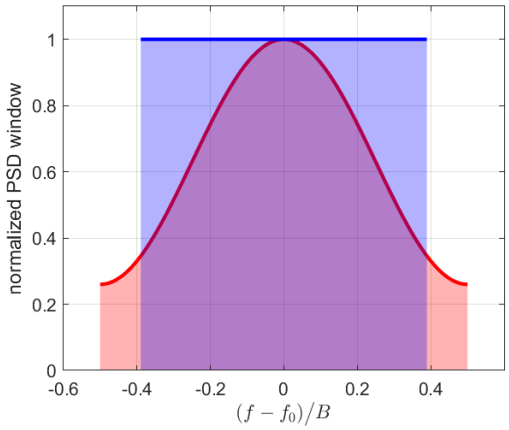


Fig. 6. Illustration of spectrum window and equivalent bandwidth. Red is the power spectral density (PSD) of windowed spectrum and blue is the PSD of the equivalent spectrum.

The equivalent bandwidths are smaller than the signal bandwidths, and their ratios can be expressed as follows:

$$\begin{aligned} \alpha_{1,\mathrm{eq}} &= \frac{B_{1,\mathrm{eq}}}{B_1} \\ \alpha_{2,\mathrm{eq}} &= \frac{B_{2,\mathrm{eq}}}{B_2}. \end{aligned} \tag{19}$$

It is clear that the equivalent frequency-domain matrix has a smaller size, which can be expressed as $L_{1,\mathrm{eq}} = N_1 \alpha_{1,\mathrm{eq}} B_1 / f_{s1}$ (azimuth) and $L_{2,\mathrm{eq}} = N_2 \alpha_{2,\mathrm{eq}} B_2 / f_{s2}$ (range). Accordingly, the equivalent variance can be computed using $\sigma_{\tilde{S}_{\mathrm{eq}}}^2 = \|\mathbf{I}\|_F^2/(L_{1,\mathrm{eq}} L_{2,\mathrm{eq}} - 1)$.

The distribution of the largest eigenvalue of the original image can be approximated using the size and variance of this equivalent matrix. The resulting p.d.f. can be expressed as follows:

$$x \triangleq \frac{\lambda_1/\sigma_{\tilde{\mathbf{S}}_{\mathrm{eq}}}^2 - c_{\mathrm{eq}}}{d_{\mathrm{eq}}} \sim \mathrm{TW}_2 \tag{20}$$

where

$$\begin{aligned} c_{\mathrm{eq}} &\triangleq \left(\sqrt{L_{1,\mathrm{eq}}} + \sqrt{L_{2,\mathrm{eq}}}\right)^2 \\ d_{\mathrm{eq}} &\triangleq \sqrt{c_{\mathrm{eq}}}\left(\frac{1}{\sqrt{L_{1,\mathrm{eq}}}} + \frac{1}{\sqrt{L_{2,\mathrm{eq}}}}\right)^{1/3}. \end{aligned} \tag{21}$$

C. Corrected Threshold for Lambda-1 Detector Under Given False Alarm Rate

Based on the above distribution, we can obtain the equation for determining the corrected threshold of the largest eigenvalue, and the resulting expression is given by

$$\lambda_{\mathrm{th}} = \left(\left(F_{\mathrm{TW}_2}^{-1}(1 - P_{\mathrm{fa}}; k, \theta, \alpha) - \alpha\right) d_{\mathrm{eq}} - c_{\mathrm{eq}}\right)\sigma_{\tilde{S}_{\mathrm{eq}}}^2. \tag{22}$$

Some illustrative examples of the calculated corrected threshold under different upsampling ratios are provided in Fig. 7, which shows that the corrected threshold is above the largest eigenvalues of clean images under different upsampling ratios and the threshold is also not too large so that the detection performance can be guaranteed.

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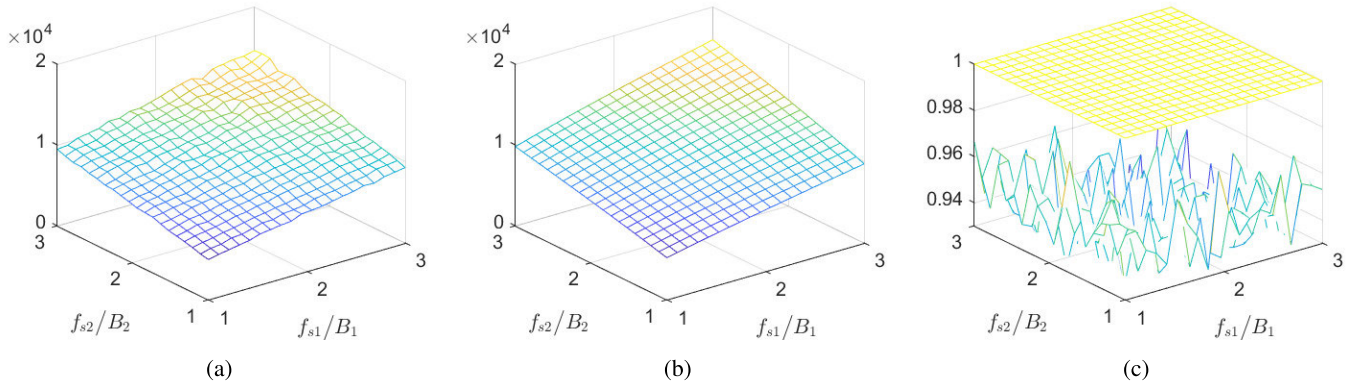


Fig. 7. Impacts of upsampling ratios to the largest eigenvalue of a clean homogeneous image and the threshold calculated using (22). f_{s1}/B_1 is the azimuth upsampling ratio and f_{s2}/B_2 is the range upsampling ratio. (a) Largest eigenvalues under different upsampling ratios. (b) Thresholds under different upsampling ratios. (c) Ratio between the largest eigenvalues and thresholds.

The above derivation assumes that the observed scene is homogeneous and its image pixels satisfy the i.i.d. Gaussian distribution. However, ideal homogeneous scenes almost do not exist in practical scenarios, and accordingly, actual SAR images will deviate from Gaussian distribution to various degrees. This will cause the actual largest eigenvalues being larger than those under the ideal conditions, causing increased false alarms. To alleviate this problem, we introduce an additional multiplicative hyperparameter $\xi \geq 1$, and use $\xi\lambda_{\text{th}}$ as the actual detection threshold.

V. Implementation Details

In this section, we provide a detailed description of our proposed method for interference detection. We first introduce a preprocessing step to improve the robustness of our method by removing outliers. Then, we describe the integrated block adaptive processing flow that combines interference detection and filtering. This flow includes segmenting the image into blocks, preprocessing each block to remove outliers, detecting interference in each block using our detection method, and finally performing interference suppression using our robust block subspace filtering method. We will also perform complexity analysis for the proposed method.

A. Improving Robustness via Outlier Removal Preprocessing

Our proposed method assumes that the image content is homogeneous, which will not be satisfied when strong scatterers exist. Strong scatterers are outliers in the homogeneous background and will cause quite large eigenvalues even if the image does not have any interference. This behavior will significantly raise the false alarm rate, as we found during this study. In order to overcome this problem, we propose a preprocessing method to remove the outliers before applying our detection method. Specifically, we use 2-D CFAR to adaptively detect strong point scatterers, and then mask them out for subsequent detection. This process can be expressed using the following expression:

$$\mathbf{S}_{\text{masked}} = (1 - \mathbf{M}) \circ \mathbf{S} \tag{23}$$

where $\mathbf{M}$ is the binary mask matrix and $\circ$ is the element-wise product. Here, $M[n, m] = 1$ indicates that $I[n, m]$ contains a strong scatterer and $M[n, m] = 0$ indicates that $I[n, m]$ is a background pixel. In our study, we use our previous proposed fast 2-D CFAR algorithm [28] to obtain the mask matrix.

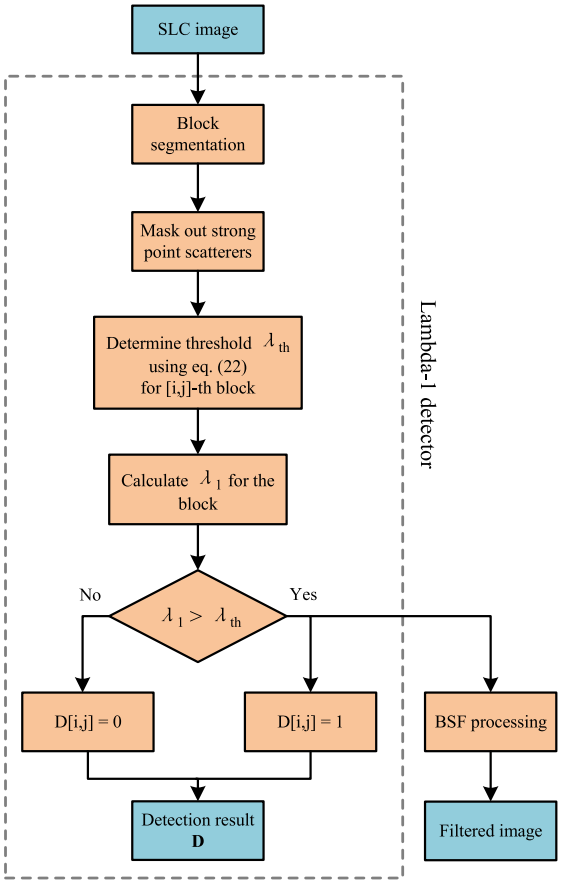


Fig. 8. Proposed Lambda-1-BSF processing flow. The Lambda-1 detector generates a binary matrix $\mathbf{D}$ as the detection result, and the BSF generates a filtered image with interference being removed.

B. Integrated Block Adaptive Processing Flow of Interference Detection and Filtering

The proposed detection method can be integrated with our previously proposed robust block subspace filtering

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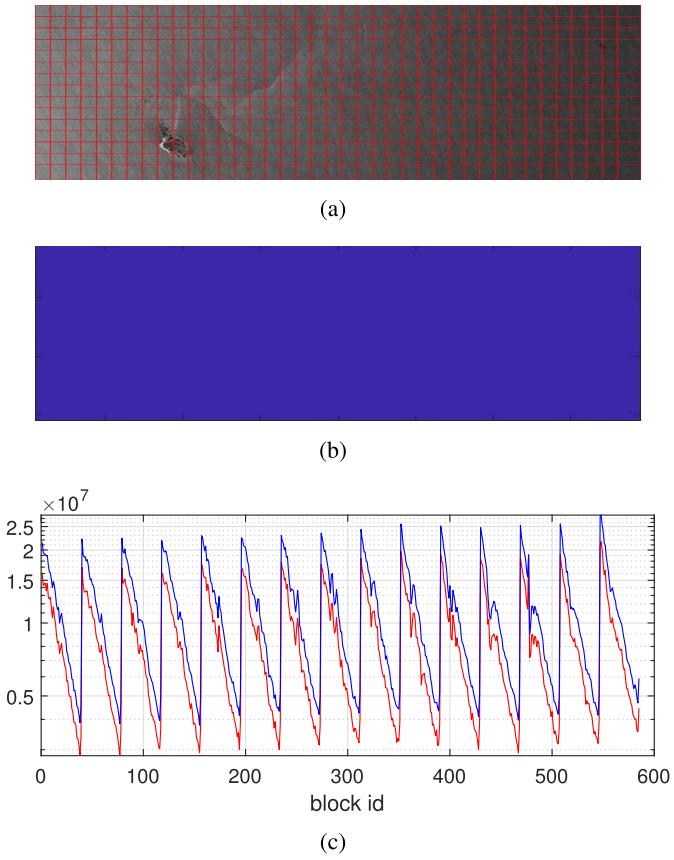


Fig. 9. Experimental image 1 and its detection result (no interference detected). (a) Experimental image. (b) Detection results. (c) 1-D plot of the largest eigenvalues (red curve) and detection threshold (blue curve) for all blocks. The block id for (p, q)th block is $(p-1)Q+q$, where $Q = \lfloor (N_r/L) \rfloor$.

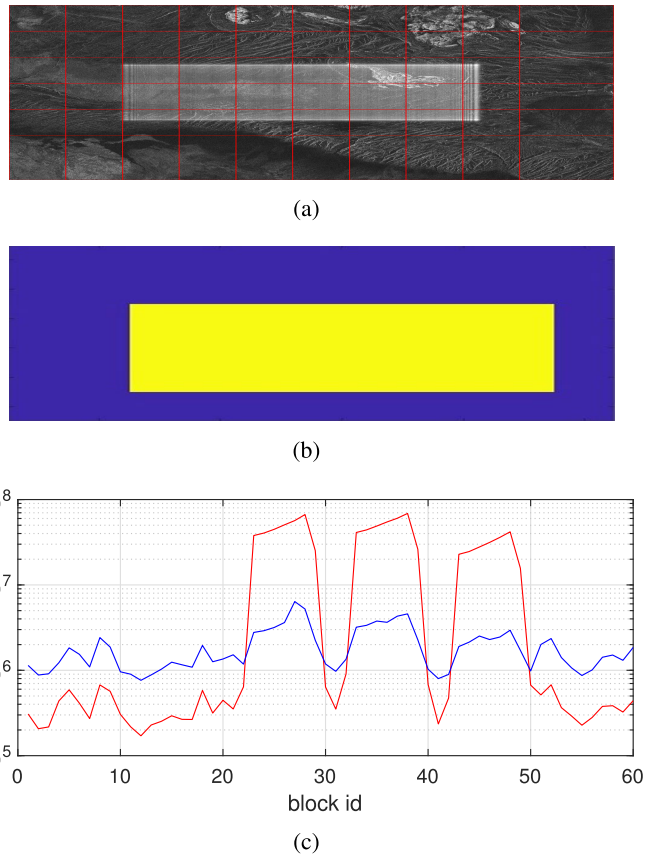


Fig. 10. Experimental image 2 and its detection result. (a) Experimental image, (b) detection results, and (c) 1-D plot of the largest eigenvalues (red curve) and detection threshold (blue curve) for all blocks.

method [26], [29] for joint interference detection and removal. The integrated processing flow can be described as the following adaptive steps.

1) *Step 1:* Segment the full $N_a \times N_r$ SLC images into $\lfloor (N_a/L) \rfloor \times \lfloor (N_r/L) \rfloor$ blocks with a bock size of $L \times L$ pixels. Here, the mathematical symbol $\lfloor (x) \rfloor$ denotes the floor function, where x is the number being rounded down to the nearest integer less than or equal to x. If the image size is not integral multiples of the block size, the bottom and right edge blocks will be assigned with a block size larger than $N \times N$.
2) *Step 2:* For each block, perform preprocessing to mask out strong outlier scatterers using 2-D CFAR in the image domain. The resulting image block is denoted by $\mathbf{S}_{\text{masked}}$.
3) *Step 3:* For each preprocessed block, use the proposed detection method described in Section VI, i.e., performing binary decision via first calculating the largest eigenvalue of the current block's covariance matrix

$$\mathbf{R} = \mathbf{S}_{\text{masked}} \mathbf{S}_{\text{masked}}^{\text{H}} \tag{24}$$

and then comparing it withn the threshold defined in (22). If interferences are detected, then go to step 4 for interference removal processing.
4) *Step 4:* For a block detected with interference, perform interference suppression using our previous proposed robust BSF filter. Specifically, the filter first performs eigenvalue decomposition on the covariance matrix

$$\mathbf{R} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{U}^{\text{H}} \tag{25}$$

where $\mathbf{U}$ is a matrix with its column being the eigenvalues of $\mathbf{R}$ and $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \ldots, \sigma_N)$ is a diagonal matrix containing the eigenvalues of $\mathbf{R}$. Here, we consider that the eigenvalues are arranged in descending order. Next, the filter constructs interference subspace using K eigenvectors indexed by the K-largest eigenvalues, i.e., $\mathbf{U}_{1:K}$. Then, the interference component, denoted by $\hat{\mathbf{Y}}$, is reconstructed by projecting the masked image block into the interference subspace

$$\hat{\mathbf{Y}} = \mathbf{U}_{1:K} \mathbf{U}_{1:K}^{\text{H}} \mathbf{S}_{\text{masked}}. \tag{26}$$

Finally, an estimate of the clean image is obtained by subtracting the reconstructed interference component from the original image block, i.e.,

$$\hat{\mathbf{S}}_{\text{clean}} = \mathbf{S} - \hat{\mathbf{Y}}. \tag{27}$$

Here, we point out that other filters can also be used in step 4.

The steps 2–4 can be implemented in parallel for speeding up the algorithm. The processing diagram of the Lambda-1 detector and its integration with BSF is shown in Fig. 8.

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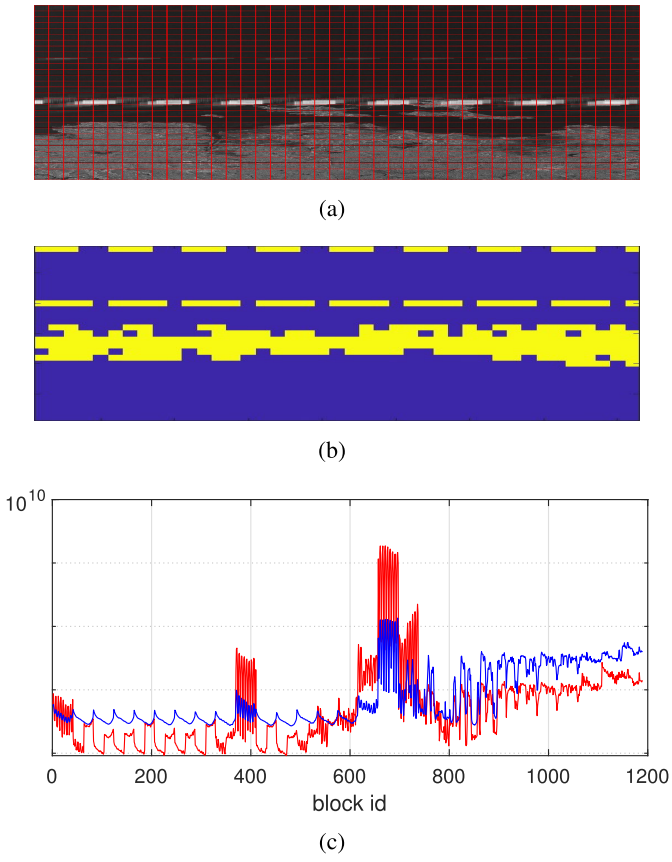


Fig. 11. Experimental image 3 and its detection result. (a) Experimental image, (b) detection results, and (c) 1-D plot of largest eigenvalues (red curve) and detection threshold (blue curve) for all blocks.

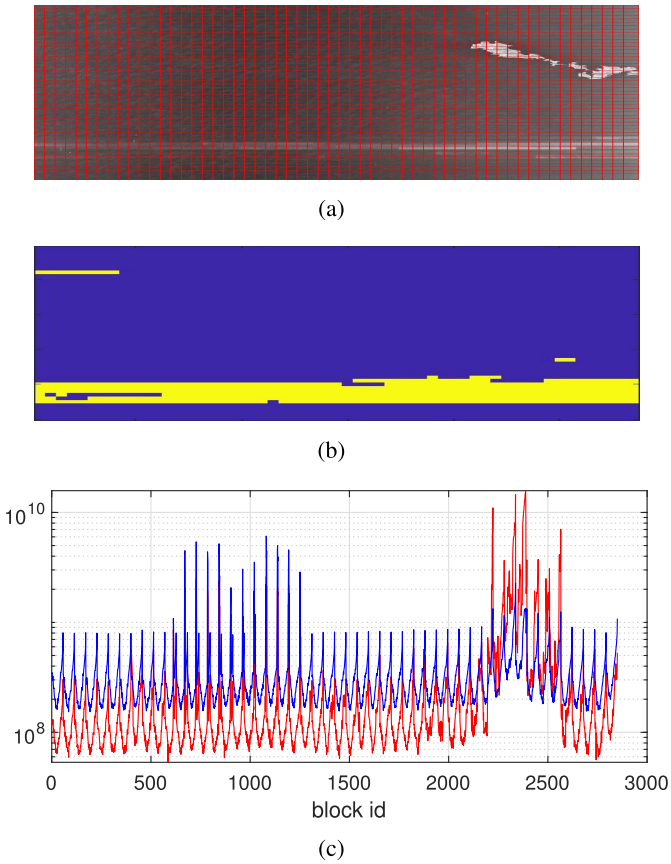


Fig. 12. Experimental image 4 and its detection result. (a) Experimental image, (b) detection results, and (c) 1-D plot of the largest eigenvalues (red curve) and detection threshold (blue curve) for all blocks.

C. Complexity Analysis

Consider that the image size is $N_a \times N_r$ and is segmented into $\lfloor (N_a/L) \times \lfloor (N_r/L)$ blocks with a block size of $L \times L$ pixels. Employing the power iteration method for calculating the largest eigenvalues and the number of iterations is set to N_{iter}, then the complexity of each block is $L^3 + L^2 N_{\text{iter}}$. Therefore, the total complexity of the detection method for the full image is $\mathcal{O}(N_a N_r L)$.

VI. Experiments

In this section, we provide real-data experiments to test the effectiveness of the proposed interference detection method. We will use two Sentinel-1 interferometric wide swath SLC images (experiments 1 and 3), a Sentinel-1 stripmap SLC image (experiment #2), and a Gaofen-3 stripmap SLC image (experiment #4). We will show how our method successfully detects interference through visualizing the largest eigenvalues and thresholds. We will also study P_d and P_{fa} curves to demonstrate the promising performance of the proposed method.

A. Experiment 1: Sentinel-1 Clean Image

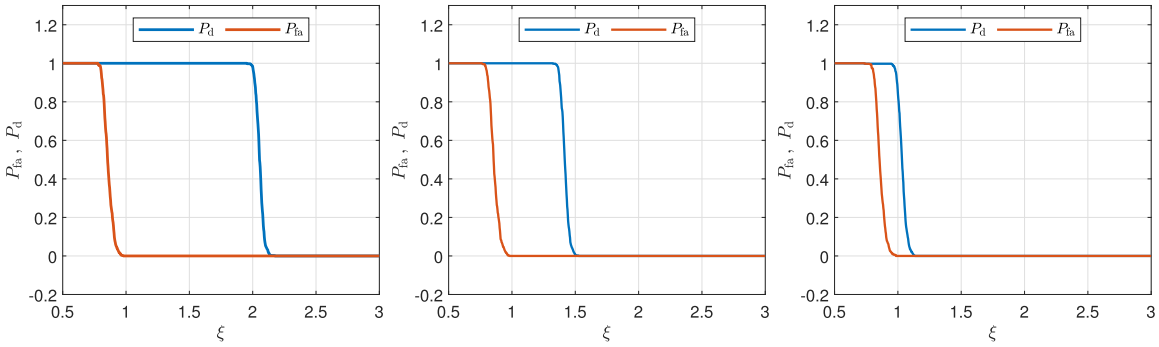
In the first experiment, we use a clean image that does not have interference to test the proposed method. The image consists of 7699 × 21 021 pixels. Our method uses a block size of 500 × 500 to segment the image as 15 × 39 blocks. Since the image size in both dimensions is not integral multiples of the block size, the last blocks (at the image bottom and right edges) are assigned a size larger than 500 × 500. Our method then performs interference detection block by block. Because the image does not contain interference, the detection method is expected to report no positive detection results.

The detection result is presented in Fig. 9(b), where the blue denotes no interference detected. To better visualize how our method determines the presence or absence of interference in each block, we plot the calculated largest eigenvalues and their corresponding detection thresholds in Fig. 9(c). In this subfigure, the largest eigenvalues and detection thresholds have sizes of 15 × 39, we reshape them (by stacking column by column) as vectors for plotting them as curves, where the red curve denotes the largest eigenvalues and the blue curves denote the thresholds. It can be seen that the largest eigenvalues vary significantly, and this is the result of two facts: 1) the mean intensity varies across different blocks and 2) the edge blocks have larger block sizes. The detection threshold produced by our method well adapts to the change in block size, and successfully determines that there is no interference.

B. Experiment 2: Sentinel-1 Image With Interference

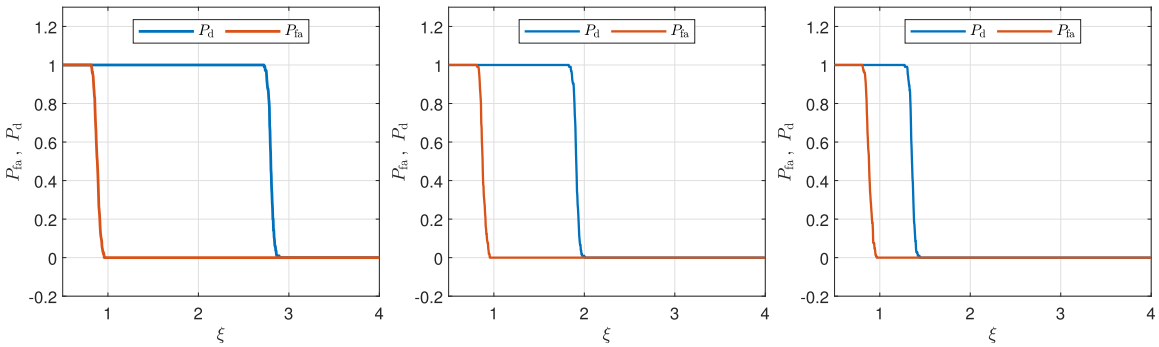
In the second experiment, we use a Sentinel-1 image with a strong LFM pulse interference to test the proposed method.

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(a) (b) (c)

Fig. 13. $P_{\rm fa}$ and P_d obtained under different threshold levels $\xi\lambda_{\rm th}$ as defined in Section IV-C. The ISR settings of (a)–(c) are -17,-19, and -21 dB, respectively. The block size is 500 × 500.



(a) (b) (c)

Fig. 14. $P_{\rm fa}$ and P_d obtained under different threshold levels $\xi\lambda_{\rm th}$ as defined in Section IV-C. The ISR settings of (a)–(c) are -17,-19, and -21 dB, respectively. The block size is 1000 × 1000.

The image has a size of 3349 × 5333 pixels. We set the block size to 500 × 500, and segment the image into 6 × 10 blocks in azimuth and range, as shown in Fig. 10(a). In this image, the middle area contains clearly visible interference. In this experiment, the time cost is 10.4 s (using an I7-13700 CPU).

The detection result of this image is shown in Fig. 10(b), where the yellow represents the existence of interference. The result shows that the interference is successfully detected by our method. We show the calculated largest eigenvalues and thresholds in Fig. 10(c), from which we can see that there are many values significantly exceed the thresholds.

C. Experiment 3: Sentinel-1 Image With Interference

In the third experiment, we use a Sentinel-1 image with both strong and weak interference to test the proposed method. The image has a size of 14 950 × 21 696 pixels. Similar to the previous experiments, we set the block size of our method to 500 × 500, which segments the image into 29 × 41 blocks in azimuth and range. The image and its block segmentation are shown in Fig. 11(a). In this image, the middle area contains clearly visible interference. In addition, the top part of the image also contains weak interference but not clearly visible. It is expected that our method can detect both the strong and weak interference. In this experiment, the time cost is 32.4 s.

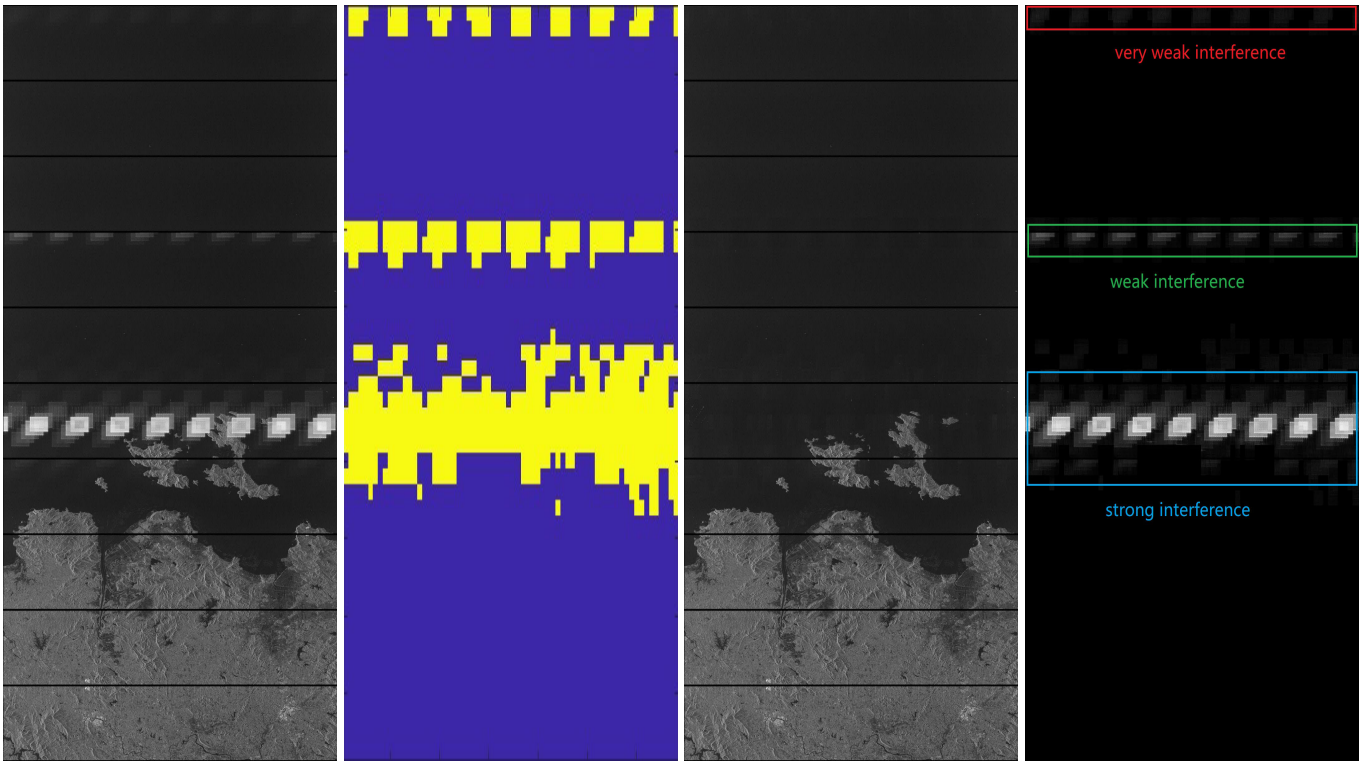
The detection result is shown in Fig. 11(b), from which we can see three horizontal areas are detected with interference. The top horizontal area contains weak interference, which are not clearly visible but also successfully detected by our method. We show the calculated largest eigenvalues and thresholds in Fig. 11(c), from which we can see that these are many values significantly exceed the thresholds due to the presence of strong interference, and some values are not only significantly large but also exceed the thresholds due to the presence of weak interference.

D. Experiment 4: Gaofen-3 Image With Interference

In the fourth experiment, we use a Gaofen-3 image with medium-level and weak interference to test the proposed method. The image has a size of 25 420 × 28 744 pixels, which segments the image into 50 × 57 blocks in azimuth and range. The image and its block segmentation are shown in Fig. 12(a). In this image, there are many areas containing pulse interference and are clearly visible, especially the image bottom. In fact, some parts of the image also contain weak interferences but we cannot identify them using our eyes. It is expected that our method can detect these weak interference. In this experiment, the time cost is 159.4 s.

The detection results are shown in Fig. 12(b). As we can see, the interference areas at the image bottom are successfully detected. We show the calculated largest eigenvalues and thresholds in Fig. 12(c), from which we can see that these values significantly exceed the thresholds. Meanwhile, there are some areas that do not have visible interference but also

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(a) (b) (c) (d)

Fig. 15. Second experimental image and its detection and filtering results. (a) Experiment #3 image, (b) detection result, (c) filtering result, and (d) removed interference.

detected to have interference, e.g., the left-top part of the image.

E. Analysis of P_{fa} and P_d

To test P_{fa}, we perform detection on $K = 420$ clean image patches $\mathbf{I}_k$, $k = 1, \ldots, K$ (extracted from the image in Experiment 1) and calculate $P_{\text{fa}} = N_1/K$, where N_1 is the number of patches detected to have interference. To obtain images for testing P_d, we add the clean image patches $\mathbf{I}_k$ scaled by a factor α to an image patch S containing interference. This results in a series of testing image patches containing interference, i.e., $\mathbf{S}_k = \alpha\mathbf{I}_k/\|\mathbf{I}_k\|_F + \mathbf{S}/\|\mathbf{S}\|_F$. The ISR of our used patch $\mathbf{S}$ is approximated to -12.9 dB, and we use $\alpha = 1.22, 1.71$, and 2.28 to set the ISR of $\mathbf{S}_k$ to $-17, -19$, and -21 dB, respectively. Then, we perform detection for $K = 420$ patches and calculate $P_d = N_2/K$, where N_2 is the number of patches detected to have interference. We use different threshold scale factor values ξ to obtain different P_{fa} and P_d. To analyze the influence of block size, we use two block sizes, i.e., 500×500 and 1000×1000, and the corresponding results are presented in Figs. 13 and. 14, respectively. The results clearly show that for a proper selection of threshold scale factor, e.g., 1.25, the presence of interference can be effectively detected with almost no false alarms provided that the ISR is not too small (>-21 dB in our experiments). However, if the ISR is too small (e.g., <-21 dB), the interference almost cannot be detected.

F. Example of Integrated Interference Detection and BSF Suppression

Fig. 15 shows the interference detection and removal results using the processing flow mentioned in Section V-B. The experimental image is the same one used in the second experiment. In this figure, (c) is the filtered image, and (d) is the removed interference. In (d), we can find three interference areas, containing very weak interference (top area), weak interference (middle area), and strong interference (bottom area). The results demonstrate that the proposed processing flow can detect and remove both strong and weak interference from SLC SAR images.

VII. Conclusion

In this article, we introduced the Lambda-1 detector for adaptive interference detection in focused SAR images. The proposed method effectively captures the increased eigenvalue caused by interference, and performs interference detection via comparing the largest eigenvalue λ_1 of image data covariance to a threshold. The threshold is adaptively determined by the TW_2 distribution under the assumption that the image pixels are homogeneous and follow an uncorrelated Gaussian distribution. This assumption is not valid for SAR images having: 1) upsampling and spectral windowing and 2) outliers (strong point scatterers). The first problem is addressed by modeling and correcting the impact of upsampling and windowing, and the second one is solved by incorporating outlier removal preprocessing, which significantly improves the algorithm's

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robustness. Experimental results demonstrate the effectiveness of the proposed method in detecting both strong and weak interferences in various SAR images. The method's advantages include its computational efficiency, robustness to outliers, and flexibility in handling different types of interference. The proposed detector requires SLC images and thus cannot detect interference from amplitude images. Although the Lambda-1 detector is designed for detecting unintentional interference, it also has the potential of detecting some jammer signals in SAR data.

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Huizhang Yang (Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in information and communication engineering from Nanjing University of Science and Technology, Nanjing, China, in 2014 and 2021, respectively.

He is currently a Post-Doctoral Researcher with Tsinghua University, Beijing, China. His research interests include radar signal processing, and synthetic aperture radar (SAR), and signal and image processing.



Ping Lang (Member, IEEE) was born in 1992. He received the B.E. degree in machine design engineering from the North University of China, Taiyuan, China, in 2015, and the M.E. degree in machine science and technology and the Ph.D. degree in signal and information processing from Beijing Institute of Technology, Beijing, China, in 2018 and 2022, respectively.

He was a Visiting Ph.D. Student with the School of Computing, University of Leeds, Leeds, U.K., from July 2019 to October 2019. He is currently a Post-Doctoral Researcher in signal and information processing with the Department of Electronic Engineering, Tsinghua University, Beijing. His research interests focus on the interdiscipline of radar signal processing and machine learning, including explainable machine learning algorithm, radar target detection and recognition, and radar system design.

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Yaomin He received the B.S. and M.S. degrees in control science and engineering from Xi'an High-Tech Research Institute, Xi'an, China, in 2017 and 2019, respectively. He is currently pursuing the Ph.D. degree in information and communication engineering with Tsinghua University, Beijing, China.

His research interests include radar electronic counter-countermeasures (ECCM), polarization signal processing, and synthetic aperture radar (SAR) signal.



Xingyu Lu (Member, IEEE) was born in Jiangsu, China. He received the B.S. and Ph.D. degrees from Nanjing University of Science and Technology, Nanjing, China, in 2014 and 2019, respectively.

In 2018, he joined the Department of Electrical and Computer Engineering, National University of Singapore, Singapore, as a Visiting Student. He is currently an Associate Professor with the Department of Electronic Engineering, School of Electronic and Optical Engineering, Nanjing University of Science and Technology. His main research interests include interference suppression for synthetic aperture radar (SAR), noise radar, integrated jamming, and detection.



Zhong Liu (Member, IEEE) received the B.S. degree in electrical engineering from Anhui University, Hefei, China, in 1983, and the M.S. and Ph.D. degrees in electrical engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 1985 and 1988, respectively.

Since 1989, he has been with the Department of Electronic Engineering, Nanjing University of Science and Technology, Nanjing, China, where he is currently a Professor. He was a Post-Doctoral Fellow with Kyoto University, Kyoto, Japan, from 1991 to 1993. He was a Researcher with The Chinese University of Hong Kong, Hong Kong, from 1997 to 1998. His research interests include chaos and information dynamics, signal processing, and radar and communication technologies.



Jian Yang (Senior Member, IEEE) received the B.S. and M.S. degrees from Northwestern Polytechnical University, Xi'an, China, in 1985 and 1990, respectively, and the Ph.D. degree from Niigata University, Niigata, Japan, in 1999.

In 1985, he joined the Department of Applied Mathematics, Northwestern Polytechnical University, Xi'an. From 1999 to 2000, he was an Assistant Professor with Niigata University. In April 2000, he joined the Department of Electronic Engineering, Tsinghua University, Beijing, China, and he was promoted to a Full Professor in 2002. He has published more than 300 articles and received many awards. His research interesting areas include radar polarimetry, feature extraction, target detection, and target classification.

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